Reconstructing the Clebsch code and its golden orientation from its deep-hole syndrome locus

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From deep holes, the cubic ***takes shape****, finds its bearings, and stands fixed while its shadows move.*

Abstract

Let A be a six-arc in $\mathrm{PG}(2, 11)$, and let $\mathcal{U}(A)$ be the projective points lying on no chord of A. For the associated $[6, 3, 4]_{11}$ MDS code, $\mathcal{U}(A)$ is its projective deep-hole syndrome locus. We prove that $\mathcal{U}(A)$ lies on a conic if and only if A is projectively equivalent to the Clebsch hexagon; in that case $\mathcal{U}(A)$ is exactly a nonsingular conic. Thus nearest-codeword data reconstruct the non-GRS Clebsch code up to monomial equivalence. The reconstructed conic then determines its polarity, and Dye's theorem identifies the stabilizer as A_5; none of this geometry is assumed.

Nearest-codeword ambiguity then reconstructs an unordered orientation torsor on six axes. Its signed orbital operator satisfies $B^2 = 5I$, and triangle holonomy gives the support cubic through $c_{ijk} = B_{ij}B_{jk}B_{ki}$; equivalently, this cubic is the sole nonsymmetric term in the diagonal determinant pencil of B. Thus the same syndrome and support data recover both the code and its golden orientation.

The proof uses a universal chord-defect identity and a partial-cover bound for rigidity, then decoder ambiguity and the orbital pentagon for orientation. As a secondary uniform consequence, any k-arc whose uncovered locus is a nonsingular conic has q odd and lies in the field window $2k - 3 \leq q \leq (k(k - 1) + 3)/3$.

Keywords. Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic.

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1 Introduction

The complete nearest-codeword structure of a code can retain geometry that is absent from its definition. The example here is a non-GRS $[6, 3, 4]_{11}$ MDS code. Its maximum-distance syndrome directions form a conic, and that coarse locus already determines the projective code: its six parity-check columns are projectively equivalent to the Clebsch hexagon, with the associated polarity and A_5 symmetry. No conic, polarity, group action, or classical hexagon structure is assumed. Thus the problem is an inverse one: when does the projective deep-hole locus determine an MDS code, and how much of its geometry can nearest-codeword data recover?

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H. For a received word v, write $\sigma(v) = Hv^{\mathsf{T}}$ and

$$w(s) = \min\{\mathrm{wt}(e) : He^{\mathsf{T}} = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ. The syndrome s specifies one coset of C, whose coset leaders are its weight-$w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\mathrm{proj}}(C) = \{[s] \in \mathrm{PG}(n-k-1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q-1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n-k$, the parity-check columns form a projective arc in $\mathrm{PG}(r-1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r-1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write $\mathcal{U}(A)$ for this extension locus. If $\mathcal{U}(A) \neq \emptyset$, then the associated code has covering radius three and $\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C)$. This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [8]; their Theorem 6.3 supplies the leader count used in Section 6. See also Hirschfeld [14] for the plane arc/covering-radius setting. We call A *conic-filling* when $\mathcal{U}(A)$ is the full $\mathbb{F}_q$-rational point set of a nonsingular conic.

Here and in Theorem 1.1, a *conic* is the projective zero set of a nonzero homogeneous quadratic form; it may be degenerate unless stated otherwise.

Theorem 1.1 (Rigidity theorem)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the following are equivalent:*

(i) the uncovered locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) A is $\mathrm{PGL}(3, 11)$-equivalent to the Clebsch hexagon.

More precisely, fix a nonsingular conic $\mathcal{C}$. The six-arcs satisfying $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_{11})$ form one orbit under $\mathrm{Stab}_{\mathrm{PGL}(3,11)}(\mathcal{C})$. Indeed, a projectivity between two such arcs carries one uncovered locus to the other. Its image of $\mathcal{C}$ therefore shares all twelve rational points with $\mathcal{C}$, so Bézout's theorem makes the two conics equal. This is the exact geometric sense in which the syndrome locus reconstructs the six-column realization.

The proof reconstructs the hexagon from conic containment alone. A line bound, a chord-defect identity, and Dye's Brianchon theorem give the conceptual implication. The same geometry then gives an exact distance oracle and all nearest-codeword multiplicities; the resulting ambiguity strata reconstruct the Brianchon points, the self-polar triangles, and the invariant support bipartition. On the reconstructed twelve-point syndrome locus, the two five-valent orbitals recover a signed operator on the six antipodal axes. Triangle holonomy identifies its switching class with the support bipartition and recovers the integral order $\mathbf{Z}[\sqrt{5}]$.

This orientation is the second structural pillar. With the normalization of Theorem 8.1, its signed operator and support cubic satisfy

$$B^2 = 5I, \qquad c_{ijk} = B_{ij}B_{jk}B_{ki},$$
$$\det(B + \mathrm{diag}(x_0, \ldots, x_5)) = e_6(x) - e_4(x) + 5e_2(x) - 125 - 2C(x).$$

Thus triangle holonomy and the diagonal determinant pencil recover the same orientation torsor directly from nearest-codeword data.

A secondary general theorem gives a uniform field window for conic-filling arcs. For an arbitrary conic-filling k-arc, the chord moments and a partial-cover theorem force q odd and $2k-3 \le q \le (k(k-1)+3)/3 < \binom{k}{2}$. For a Clebsch hexagon K, the same chord-defect identity gives $|\mathcal{U}(K)| = q^2 - 14q + 45$, with off-conic excess $(q-4)(q-11)$ when $q \equiv 3 \pmod 4$.

The configuration goes back to Clebsch [7, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [12, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [11, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [25]. Deep holes of Reed–Solomon codes have both an algorithmic history [13, 6] and a geometric classification in the projective case [27]. Kaipa makes the deep-hole/MDS-extension equivalence explicit for Reed–Solomon codes [16], while Wu, Ding, and Chen formulate a corresponding extension criterion for general MDS codes and dual deep holes [26]. Those results begin with a fixed code and characterize its extensions. The inverse statement here instead reconstructs a non-GRS projective code from its entire maximum-distance syndrome locus.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [7, 12, 11, 4, 25]. We prove the conic-locus identification of Proposition 6.1 directly here. The new contribution is the symmetry-free coding criterion, the decoder reconstruction and support bipartition, and the uniform conic-filling window. The reconstructed support bipartition and syndrome cover also determine a balanced signed two-graph and its golden operator; their triangle products recover a cubic determinant line. The underlying six-nodal S_5-symmetric cubic is known [5]; what is proved here is its intrinsic recovery from the code's syndrome and support data, together with the integral continuation order. A separate computational companion [19] proves sharper finite consequences: a degree-three characterization and numerical gap at $q = 11$, uniqueness of $q = 11$ among conic-filling six-arcs, classification through eight points, and the original computational form of an exact $q = 13$ binary incidence-code theorem. Paper IV gives the structural and reproducible account of that $q = 13$ consequence [23].

This is Paper I of a four-paper sequence organized by what successive constructions forget and recover. Paper II studies what the conic matching quotient remembers [20]. Paper III, also called *passages*, supplies the characteristic-zero golden source and develops its broader structural consequences [22]. The reconstructed operator also has uses beyond this inverse problem: the unnumbered Golden quantum-statistics companion studies its exchange-statistical interpretation [21].

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}: \; XZ = Y^2$$

in $\mathrm{PG}(2, 11)$, with its 12 points identified with $\mathrm{PG}(1, 11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3, 11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

A convenient representative is

$$A = \{(1, 10, 0), \; (1, 9, 1), \; (1, 4, 7), \; (1, 8, 5), \; (0, 1, 4), \; (1, 1, 7)\}.$$

It is a six-arc disjoint from $\mathcal{C}$; its fifteen joins also avoid $\mathcal{C}$. The following invariant construction explains its symmetry.

Definition 2.1 (Clebsch hexagon). Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The six Sylow 5-subgroups of A_5 each fix two points of $\mathcal{C}$. The *Clebsch hexagon* is the set of poles of the six chords joining these fixed pairs. More generally, following Dye, a Clebsch hexagon over a field is a six-arc with exactly ten nonvertex triple-chord concurrence points, called its Brianchon points.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an *exterior*, or *passant*, line is disjoint from the conic. Edge's six vertices are external, their fifteen joins are exterior, and those joins concur three at a time at ten internal Brianchon points [12, §§29–32]. Thus the hexagon is a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [4, pp. 143,146]. Their $q = 11$ list also contains a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. Thus exteriority alone gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; the arc hypothesis selects the Clebsch branch, and Proposition 6.1 proves equality.

The displayed hexagon is the arc K_2 of Storme and Van Maldeghem [25, Props. 11–12] and Dye's synthetic hexagon when 5 is a square in the base field [11, Theorem 1, pp. 275–276]. In characteristic 11, Dye's Theorems 1–3 give projective uniqueness, the unique associated polarity, and stabilizer A_5; in the displayed coordinates the associated conic is $\mathcal{C}$ [11, pp. 275–278]. His calculation also shows that the six vertices lie on no conic in characteristic 11 [11, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem record that the arc is incomplete at $q = 11$ [25, Prop. 13], so the associated redundancy-three code has covering radius three.

3 A universal chord-defect identity

Theorem 3.1 (Universal chord defect). *Let A be a k-arc in $\mathrm{PG}(2,q)$ with $k \geq 4$. Let n_i count the off-arc points incident with exactly i chords, put*

$$r = \lfloor k/2 \rfloor, \qquad t(A) = \sum_{i=3}^{r} \binom{i-1}{2} n_i,$$

and write $m = \binom{k}{2}$. Then

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3\binom{k}{4} - k + 1\right) - t(A),$$
$$0 \leq t(A) \leq 3\binom{k}{4}\frac{r-2}{r}, \qquad k \leq 5 \Longrightarrow t(A) = 0.$$

Proof. At a point off A, concurrent chords have pairwise disjoint endpoint pairs, so at most r chords occur. Counting chord–point incidences and pairs of disjoint chords gives

$$\sum_i i n_i = m(q-1), \qquad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

Since $\binom{i}{2} = (i-1) + \binom{i-1}{2}$, the number of covered off-arc points is $m(q-1) - 3\binom{k}{4} + t(A)$. Subtraction from the $q^2 + q + 1 - k$ off-arc points proves the identity. Finally,

$$\binom{i-1}{2} \leq \frac{r-2}{r}\binom{i}{2} \qquad (3 \leq i \leq r),$$

and the second moment gives the defect bound. $\square$

4 The rigidity theorem

Proposition 6.1 concerns a specific arc. We now prove the symmetry-free characterization stated in Theorem 1.1. The uncovered conic determines the resulting Clebsch hexagon's polarity, and Dye's stabilizer theorem then identifies its projective stabilizer as A_5; neither is assumed.

Conic containment. The condition is that the uncovered locus $\mathcal{U}(A)$ lies on a conic, not that the arc A itself does. The proof below uses only the former condition and imposes no hypothesis that the six vertices lie on a conic.

We first isolate the geometric input needed to exclude degenerate containing conics.

The exterior-line case is a six-point instance of the nonexistence in odd order of nontrivial *hyperfocused arcs*, whose secants determine only $k-1$ points on an exterior line [2, 9, 18]. We give a direct proof because the three possible line positions are then handled together.

Lemma 4.1 (Line bound)**.** *Let A be a six-arc in* $\mathrm{PG}(2,q)$*, where q is odd. Every line contains at most $q-5$ points of $\mathcal{U}(A)$.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points. Indeed, two of those chords sharing an endpoint meet only at that endpoint, which lies off ℓ; their intersections with ℓ therefore give a proper edge-colouring of K_5, whose chromatic index is five. Together with the vertex, at least six of the $q+1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \varnothing$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them: chords sharing an endpoint meet only at that arc vertex, not on ℓ, so all chords through one point of ℓ have disjoint endpoint pairs. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6, necessarily a one-factorization. The one-factorization of K_6 is unique up to isomorphism [17]; after relabelling, three factors are

$$01|23|45, \qquad 02|14|35, \qquad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0=(0,0)$, $p_1=(x,0)$ and $p_2=(0,y)$, where $xy\neq 0$. Writing $p_3=(a,y)$, the first two factors force $p_4=(x,b)$ and $p_5=(a,b)$. Parallelism of 03 with 15 and with 24 gives

$$ab-ay+xy=0, \qquad ab-ay-xy=0.$$

Their difference is $2xy=0$, impossible for odd q. Thus a disjoint line has at least six covered points, and in all cases at most $(q+1)-6=q-5$ points of $\mathcal{U}(A)$ remain. $\square$

Proof of Theorem 1.1. Suppose that $\mathcal{U}(A)$ lies on a conic Q. If Q is nonsingular, then $|\mathcal{U}(A)|\leq 12$. If it is degenerate, its rational points lie on at most two rational lines, apart from the nonsplit rank-two case in which only the singular point is rational. Lemma 4.1 again gives $|\mathcal{U}(A)|\leq 12$.

Theorem 3.1, on the other hand, gives

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ counts the nonvertex triple-chord concurrences, namely Dye's Brianchon points. Dye proves $c(A)\leq 10$, with the equality cases, when they occur, forming one $\mathrm{PGL}_3(K)$-orbit [11, §2.2–2.3, Theorem 1, pp. 275–276]. Here $\operatorname{char}\mathbb{F}_{11}=11\neq 2,5$ and $5=4^2$ in $\mathbb{F}_{11}$, so Dye's equality orbit exists. Thus $|\mathcal{U}(A)| \geq 12$. Both bounds are equalities: $c(A) = 10$, so A is a Clebsch hexagon.

For a Clebsch hexagon the fifteen chords are exterior to Dye's associated nonsingular conic $\mathcal{C}$. Hence $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; both sets have twelve points by the preceding equality, so they coincide. The original Q cannot be degenerate, since Bézout bounds its intersection with $\mathcal{C}$ by four. This proves (i)⇒(iii), and the same exteriority and count give (iii)⇒(ii)⇒(i). Dye's stabilizer theorem identifies the stabilizer with A_5. □

Projective equivalence of the unordered parity-check column sets is equivalent to monomial equivalence of the associated codes: choose column representatives after the projectivity, then absorb their nonzero scalar factors into coordinate scalings. Thus Theorem 1.1 also proves the coding formulation in the abstract.

Remark 4.2 (Why this rigidity is not in the extension-locus literature). The set $\mathcal{U}(A)$ is the classical extension locus, studied for sufficiently large arcs through Segre's tangent-envelope method; see Ball and Lavrauw [1, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime and do not record the conic rigidity detected here.

The same counting argument gives a uniform field window once the uncovered locus fills a conic.

Theorem 4.3 (Conic-filling window and six-arcs)**.** *If $|\mathcal{U}(A)| = q + 1$, then*

$$t(A) = q^2 - mq + m + 3\binom{k}{4} - k, \qquad q^2 - mq + m - k + \frac{6}{r}\binom{k}{4} \leq 0,$$

and $q < m$. Under the same hypothesis $|\mathcal{U}(A)| = q + 1$, if q is even, then $\mathcal{U}(A)$ is not an arc. Hence if $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and

$$2k - 3 \leq q \leq \frac{k(k-1)+3}{3} < \binom{k}{2}.$$

For $k = 6$, writing $c(A) = n_3$ gives

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad 0 \leq c(A) \leq 15, \qquad c(A) = (q-6)(q-9)$$

when $|\mathcal{U}(A)| = q + 1$. For the displayed Clebsch hexagon, Dye's ten Brianchon points give $c(A) = 10$; its fifteen chords have 45 disjoint pairs, cover 115 of the 127 off-arc points, and leave exactly twelve uncovered.

Proof. The first formula is Theorem 3.1 rearranged, and its comparison with the defect bound gives the quadratic inequality. There are $q^2 - k$ covered off-arc points, so $q^2 - k \leq m(q-1)$; if $q \geq m$, then $0 \leq q(q-m) \leq k - m < 0$, a contradiction.

Suppose q is even and $\mathcal{U}(A)$ is an arc. Since it has $q+1$ points, it has a nucleus $N \notin \mathcal{U}(A)$ through which every line is tangent [14, p. 177]. Every chord of A is disjoint from $\mathcal{U}(A)$, so no chord contains N; nor can N be a vertex. Thus $N \in \mathcal{U}(A)$, a contradiction.

In the conic case, q is therefore odd and every chord is passant. An off-conic point lies on at most $(q+1)/2$ passants, so the $k-1$ chords through a vertex give $q \geq 2k-3$.

The m chords cover the complement of the conic by passant (elliptic) lines. The exact conic-complement bound of Blokhuis, Brouwer, and Szőnyi gives [3, Proposition 1.6]

$$m \geq 2q - 1 - \frac{q+1}{2} = \frac{3(q-1)}{2}.$$

Rearranging gives $q \leq (k(k-1)+3)/3$; the final strict inequality follows from $k \geq 4$.

For $k = 6$, one has $r = 3$ and $t(A) = n_3 = c(A)$, giving the displayed specialization. For the displayed arc, Dye's Brianchon theorem gives $c(A) = 10$ [11, Theorem 1, pp. 275–276]. At $q = 11$, the two moments read $\sum in_i = 150$ and $\sum \binom{i}{2} n_i = 45$, whence $n_1 + n_2 + n_3 = 115$. $\square$

Remark 4.4 (Sharpness of the window). Both bounds are attained. A projective four-frame over $\mathbb{F}_5$ has a conic-filling uncovered locus and satisfies $q = 2k - 3 = (k(k-1) + 3)/3$. The Clebsch six-arc over $\mathbb{F}_{11}$ attains the upper bound. We do not know an infinite family attaining either bound; the computational companion [19] shows that, through $k = 8$, these are the only conic-filling examples.

5 The Clebsch family across fields

Recall that A is conic-filling when $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{Q}$. Equivalently, $A \cap \mathcal{Q}(\mathbb{F}_q) = \varnothing$, its chords cover every point outside $A \cup \mathcal{Q}(\mathbb{F}_q)$, and no chord meets $\mathcal{Q}(\mathbb{F}_q)$. For the Clebsch family, the chord-defect identity gives a direct formula for the uncovered locus and isolates the conic-filling specialization.

Proposition 5.1 (The Clebsch-family uncovered formula). *Let K be a Clebsch hexagon over $\mathbb{F}_q$; equivalently, K is a six-arc with ten Brianchon points. By Dye's existence theorem, the characteristic is not two and 5 has a square root in $\mathbb{F}_q$, possibly zero in characteristic five. Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod 4$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$ and

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q-4)(q-11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [11, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(K)$. Thus $c(K) = 10$, and Theorem 3.1 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of K are non-secants of the associated conic when $q \equiv 3 \pmod 4$ [11, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q-4)(q-11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points. $\square$

Remark 5.2 (Three Clebsch-family regimes). The polynomial $q^2 - 14q + 45 = (q-5)(q-9)$ vanishes exactly at $q = 5, 9$. The root $q = 5$ lies in characteristic five, where the associated-conic construction of Definition 2.1 is not available. At $q = 9$, however, Dye's construction applies: $5 = -1$ is a square in $\mathbb{F}_9$, and the characteristic is three. Thus the formula gives an empty uncovered locus, so the Clebsch six-arc is complete and not conic-filling. At $q = 11$ the uncovered locus is instead exactly the associated conic.

At $q = 19$, the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [11, p. 281], while Proposition 5.1 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus $120 = (19 - 4)(19 - 11)$ additional points escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at $q = 11$.

6 The code and its projective syndrome locus

Take the parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 10 & 9 & 4 & 8 & 1 & 1 \\ 0 & 1 & 7 & 5 & 4 & 7 \end{pmatrix},$$

whose columns are the six displayed points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6, 3, 4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [24].

The arc–coset dictionary from the introduction applies without a GRS hypothesis [8, Def. 6.1, Thm. 6.3]. Each uncovered direction x lifts to $q - 1$ weight-three cosets, and each has $\binom{6}{3} = 20$ minimum-weight leaders. Geometrically, the same condition says that adjoining x as a seventh column preserves the MDS property.

For example, $s = (1, 0, 0)$ lies on $\mathcal{C}$ and is therefore uncovered. The error

$$e = (7, 4, 1, 0, 0, 0)$$

satisfies $He^{\mathsf{T}} = s$. No error of weight one or two has syndrome s, while the twenty triples of coordinate positions each supports one weight-three leader. Thus this single syndrome exhibits the complete dictionary: its projective direction is a conic point, its coset has distance three, and its nearest-codeword ambiguity has size twenty.

Proposition 6.1 (The uncovered locus is the conic)**.** *The covering radius of C is three, and*

$$\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C) = \mathcal{C}(\mathbb{F}_{11}).$$

Proof. Theorem 4.3 gives $|\mathcal{U}(A)| = 12$. The exteriority of all fifteen chords gives $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$. Since a nonsingular conic has twelve $\mathbb{F}_{11}$-points, equality follows without using the orbit classification. The arc–coset dictionary identifies this set with the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three. $\square$

Corollary 6.2 (Directions, cosets, leaders, and received words)**.** *$\mathcal{U}(A)$ has 12 directions, exactly the points of $\mathcal{C}$. Above them lie $12(11-1) = 120$ nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has $\binom{6}{3} = 20$ minimum-weight leaders, for 2400 leaders in all, and contains $|C| = 11^3 = 1331$ received words. Consequently C has $120 \cdot 1331 = 159720$ received-word deep holes.*

Proof. Immediate from Proposition 6.1, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \to \mathrm{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. □

The stabilizer further organizes all the small point sets that occur in the decoder and orientation constructions.

The next proposition is a structural calculation, not an extraction from the finite orbit tables. We include the stabilizer bookkeeping because it is precisely what makes the seven-orbit decomposition independently auditable: element eigenspaces determine subgroup fixed points, orbit–stabilizer determines their masses, and the masses exhaust the projective plane.

Proposition 6.3 (The A_5 point orbits)**.** *Let $G = \mathrm{Stab}(A) \cong A_5$. Its orbits on $\mathrm{PG}(2, 11)$ have lengths*

$$6,\ 10,\ 12,\ 15,\ 30,\ 30,\ 30.$$

The first four are respectively A, the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G-orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [11, Theorems 2–3 and Corollary 1, pp. 276–279]. We derive the orbit sizes without using the finite enumeration.

The argument first determines the points with each possible nontrivial stabilizer, then applies orbit–stabilizer to recover the orbit lengths.

Since $11 \nmid 60$, Maschke's theorem applies. Moreover 5 is a square in $\mathbb{F}_{11}$, so the two ordinary ternary icosahedral characters are realized over $\mathbb{F}_{11}$, and the ordinary and Brauer characters agree on every class. On the classes $1, 2, 3, 5A, 5B$, either has values $3, -1, 0, \phi, \bar{\phi}$, where $\phi + \bar{\phi} = 1$ and $\phi\bar{\phi} = -1$; using the class sizes $1, 15, 20, 12, 12$ gives

$$\langle \chi, \chi \rangle = \frac{9 + 15 + 12(\phi^2 + \bar{\phi}^2)}{60} = 1.$$

Thus the cross-characteristic reduction is absolutely irreducible, and G fixes no projective point.

The characteristic polynomials make the elementwise fixed-point counts transparent. An involution has eigenspaces of dimensions one and two, and therefore fixes $1 + (11 + 1) = 13$ projective points. An element of order three has characteristic polynomial $(T-1)(T^2+T+1)$; the quadratic factor is irreducible over $\mathbb{F}_{11}$, so it fixes one projective point. An element of order five has three distinct eigenvalues in $\mathbb{F}_{11}$ because $5 \mid 10$, and hence fixes three projective points.

It remains only to intersect these eigenspaces inside the standard proper subgroups of A_5. Their numbers of conjugates follow from their normalizers. A D_5 retains one of the three C_5 eigenlines and exchanges the other two; an S_3 retains the unique C_3 eigenline; a V_4 has three simultaneous eigenlines; and its overgroup A_4 permutes those three and fixes none. The standard subgroup classification of A_5 shows that these exhaust the possible nontrivial point stabilizers. Here D_5 has order ten, and the last row counts points whose stabilizer is exactly the displayed subgroup type.

H	A_4	D_5	S_3	C_5	V_4	C_3
number of conjugate subgroups	5	6	10	6	5	10
fixed points for one H	0	1	1	3	3	1
points with stabilizer conjugate to H	0	6	10	12	15	0

For the third row, the two exchanged C_5 eigenlines have exact stabilizer C_5; the unique C_3 line is already fixed by its S_3 normalizer; and the three V_4 eigenlines have exact stabilizer V_4 because A_4 permutes them without fixing one. By orbit–stabilizer, the third row already gives the orbits of lengths 6, 10, 12, and 15. To account for the points with exact stabilizer C_2, observe that each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4. Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2. There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6+10+12+15+90 = 133$ exhausts the plane. Dye's Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5-stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [11, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. □

6.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^{\times} \longrightarrow \mathrm{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars: a projectivity fixing all six column rays fixes a projective frame and is the identity. The sequence splits. Since $\gcd(3, 10) = 1$, each projective class has a unique determinant-one representative. These representatives are closed under multiplication, and their induced monomial lifts are unique once the displayed column representatives are fixed; hence they give an A_5 complement. The global scalars are central, so

$$\mathrm{MAut}(C) \cong C_{10} \times A_5$$

has order 600.

Proposition 6.4 (One orbit on received-word deep holes)**.** *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C|\,|\mathrm{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^{\times}$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w, choose $m \in \mathrm{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w. Both kinds of generator are Hamming isometries preserving C. The monomial group normalizes the translation subgroup, since $m\, t_c\, m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \mathrm{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. $\square$

6.2 Decoding consequences

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters. This is the radius-three instance of the coset-weight framework of Davydov, Marcugini, and Pambianco [8, §6]; the secant indices below refine its weight-two stratum by nearest-codeword multiplicity.

The special geometry also makes distance diagnosis and nearest-codeword structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$. For a point $x \notin A$, its *secant index* is the number of chords of A through x; by the dictionary above, this is the number of weight-two leaders for each nonzero syndrome on the ray x. The following oracle is likewise derived from incidence identities; the syndrome enumeration in the formal package is an independent check of it, not an input to its proof.

Proposition 6.5 (Complete syndrome oracle and ambiguity enumerator). *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^{\mathsf{T}}$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 *nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are*

$$960, \qquad 150, \qquad 100, \qquad 120.$$

For a distance-two syndrome the number of nearest codewords is the secant index of its direction. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 6.1 identifies the last set with $Q = 0$, proving the oracle. For completeness, the ambiguity counts come from three incidence equations rather than a syndrome table. If n_i counts off-arc points of secant index i, then

$$n_0 = 12, \qquad n_1 + n_2 + n_3 = 133 - 6 - 12 = 115,$$

while chord–point and disjoint-chord-pair incidence give

$$n_1 + 2n_2 + 3n_3 = 15(11 - 1) = 150, \qquad n_2 + 3n_3 = 3\binom{6}{4} = 45.$$

Dye's ten Brianchon points give $n_3 = 10$, and hence $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. ☐

6.3 Self-polar structure

Edge's finite-plane construction [12, §§30.2–31] and Dye's general-field synthetic theory [11, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into five perfect matchings of K_6. Section 7 turns its ten pairs of matchings into the Brianchon–support dictionary.

7 The invariant support bipartition

Label the six coordinates by $0, \dots, 5$. The 20 three-element coordinate supports form two complementary A_5-orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive. We use *support bipartition* only as shorthand for this unoriented bipartition. Its ten complementary pairs carry the Petersen graph in Figure 1.

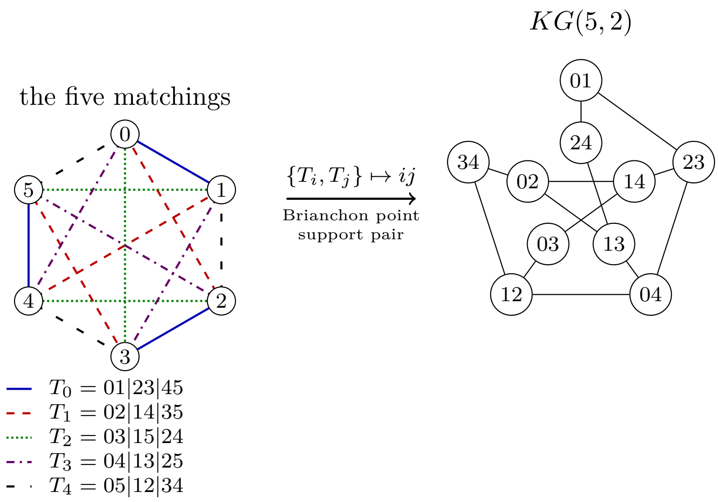

Figure 1: The five-matchings–Petersen dictionary. The line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair $\{T_i, T_j\}$, their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex ij; adjacency means that the index pairs are disjoint.

Proposition 7.1 (Brianchon–support dictionary)**.** *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [12, 11], regarded as five perfect matchings, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle. Let $\beta(T, T') = \{S, S^c\}$ be the unordered pair of alternating vertex classes of* *the cycle. For example, $T_0 \cup T_1$ is the cycle $0,1,4,5,3,2$, so $M(T_0,T_1) = 05|13|24$ and $\beta(T_0,T_1) = \{\{0,3,4\},\{1,2,5\}\}$.*

The three chords indexed by $M(T,T')$ concur at a point $b(T,T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \qquad \{T,T'\} \longmapsto b(T,T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S,S^{\mathrm{c}}\} : |S| = 3\}, \qquad \{T,T'\} \longmapsto \beta(T,T'),$$

are A_5-equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of $\mathcal{T}$. Hence $\{T,T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T,T')$, so the ten triangle pairs give exactly the ten perfect matchings outside $\mathcal{T}$. Applying the alternating-class rule to the five displayed matchings gives ten distinct complementary support pairs, hence all ten such pairs. Edge's Clebsch construction says that the three chords of each such matching concur at the ten distinct Brianchon points [12, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the five invariant matchings and incidence, both bijections and their compatibility with the support map are A_5-equivariant. □

Corollary 7.2 (The decoder reconstructs the Brianchon configuration)**.** *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five matchings in $\mathcal{T}$.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 7.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings outside $\mathcal{T}$. □

Proposition 7.3 (Invariant support bipartition)**.** *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary A_5-orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s, each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200+1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The nontrivial coset of A_5 in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 7.1, the ten complementary support pairs are canonically identified both with the two-subsets of the five self-polar triangles and with the ten Brianchon points.

Proof. The degree-six action of A_5 on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique set of five perfect matchings, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 1 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome s and a three-support S. The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s. None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A, contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S. Finally, every monomial automorphism induces a support permutation in A_5, by the exact sequence in Section 6.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve C, even after coordinate rescaling, because its support permutations lie outside that quotient A_5. $\square$

8 The intrinsic orientation two-graph

The support bipartition and the twelve deep-hole directions carry two apparently different orientation choices. We now show that they are the same datum. This comparison uses only the reconstructed A_5-sets in this paper. We use the standard correspondence between two-graphs, switching classes, and Seidel matrices; see [10, §2].

The cubic threefold that appears below agrees with a known six-nodal model. Cheltsov, Tschinkel, and Zhang classify cubics already assumed to have six nodes in general linear position and, in the S_5 case, give the equation [5, §7, equations (7.4)–(7.5), Proposition 7.3]. Choosing $x_0 = 0$ as a representative of $\mathbf{Q}^6/\mathbf{Q1}$ and interchanging their coordinates y_3 and y_5 identifies their equation (7.5) with $C = 0$ below. More explicitly, if $G(y_1, \ldots, y_5)$ denotes the left-hand side of their equation (7.5), then

$$C(0, y_1, y_2, y_3, y_4, y_5) = G(y_1, y_2, y_5, y_4, y_3).$$

The result here is not a new construction of that cubic or its abstract S_5-action. It recovers the cubic intrinsically from the syndrome locus: the support-orbit signs give its equation, and the diagonal determinant identity identifies it with the golden continuation operator. Because the cited proposition assumes six nodes, we prove singular-locus completeness independently below; the coordinate match is not used in that step.

Let $\Omega = \mathcal{C}(\mathbb{F}_{11})$. Its points have stabilizer C_5. The two points fixed by each Sylow 5-subgroup form an antipodal pair, giving an A_5-equivariant quotient

$$\Omega = A_5/C_5 \longrightarrow A_5/D_5 = \Xi$$

onto six axes. Write R for its deck involution. The stabilizer of a point of Ω has suborbits of lengths $1, 1, 5, 5$; let A and A' be the adjacency matrices of the two five-valent orbitals.

Theorem 8.1 (Support cubic and golden continuation)**.** *The syndrome locus reconstructs an unordered orientation torsor with the following equivalent presentations.*

(i) Give the two A_5-orbits of three-coordinate supports opposite signs. Their signed moments in degrees zero, one, and two vanish, while the first nonzero moment is the cubic line spanned by

$$C(x) = \sum_{i<j<k} c_{ijk} x_i x_j x_k, \qquad c_{ijk} \in \{\pm 1\}.$$

(ii) On the fibre-odd lattice

$$L^- = \{f \in \mathbf{Z}^\Omega : f(R\omega) = -f(\omega)\},$$

either five-valent orbital induces, after choosing one representative over each axis, a symmetric signed matrix B with

$$B_{ii} = 0, \qquad B_{ij} \in \{\pm 1\}, \qquad B^2 = 5I.$$

Changing the six representatives switches B to DBD, and exchanging the orbitals negates it.

The two presentations determine one another. Under the unique A_5-equivariant identification of the coordinate positions with Ξ,

$$\boxed{c_{ijk} = B_{ij}B_{jk}B_{ki}.}$$

Equivalently, if $e_d(x)$ denotes the elementary symmetric polynomial, then

$$\boxed{\det(B + \operatorname{diag}(x_0, \dots, x_5)) = e_6(x) - e_4(x) + 5e_2(x) - 125 - 2C(x).}$$

Thus the support cubic is the sole nonsymmetric term in the diagonal determinant pencil of the golden operator. In homogeneous form, with $F_B(x,z) = \det(zB + \operatorname{diag}(x))$, this becomes

$$F_B(x,z) - F_B(x,-z) = -4z^3C(x).$$

The cubic descends to the augmentation space $\mathbf{Q}^6/\mathbf{Q}\mathbf{1}$. Conversely, its two-graph identities and vanishing signed pair moments reconstruct B and force $B^2 = 5I$; there is a unique balanced switching class up to relabeling.

Corollary 8.2 (Nodes, symmetry, and integral commutant)**.** *The cubic also recovers the unlabelled axis carrier: in*

$$\mathbf{P}(\mathbf{Q}^6/\mathbf{Q}\mathbf{1})$$

its zero locus has exactly six singular points $p_a = [\mathbf{1} - 6e_a]$, all ordinary nodes. They form a projective frame, and

$$\operatorname{Aut}_{\mathrm{proj}}(C = 0) \cong S_5,$$

with the oriented-cubic stabilizer A_5 as its index-two subgroup. Consequently the continuation algebra and its integral commutant are

$$\mathbf{Q}[B] \cong \mathbf{Q}(\sqrt{5}), \qquad \operatorname{End}_{\mathbf{Z}A_5}(L^-) = \mathbf{Z}[B] \cong \mathbf{Z}[\sqrt{5}].$$

Proof of Theorem 8.1 and Corollary 8.2. The proof has four stages: the orbital pentagon produces the golden operator, triangle holonomy identifies its switching class with the support cubic, principal minors give the determinant pencil, and the cubic then recovers the node frame, its symmetry, and the integral commutant.

The orbital pentagon. The degree-six action has point stabilizer D_5. Its two orbits on three-subsets have size ten, are exchanged by complementation, and produce a nonzero signed cubic determined up to its overall sign.

Antipodal exchange carries one five-suborbit of Ω to the other, so $A' = AR$. Both orbitals are self-paired. Indeed, in the coset model $\Omega = A_5/H$ with $H = \langle r \rangle$, $r = (12345)$, the two nontrivial double cosets have representatives $a = (345)$ and $b = (23)(45)$; they are inverse-stable because $r^2ar^2 = a^{-1}$ and $b = b^{-1}$.

The stabilizer H of a point acts regularly on the other five axes. Each five-suborbit is an H-orbit, so its projection to those axes is an equivariant map between regular five-sets and hence a bijection. Thus between two fibres exactly one point lies in $A(\omega)$ and the other in $A'(\omega)$. On L^- the two orbitals act oppositely. Choosing one point above each axis therefore represents the restriction of A by a symmetric zero-diagonal matrix B with off-diagonal entries ± 1. Changing a lift switches the corresponding row and column; exchanging the orbitals negates B.

Fix ω and choose the other five lifts in $A(\omega)$, so $B_{0i} = 1$. The stabilizer C_5 of ω acts regularly on the other five axes. Their signs are therefore constant on the sides and on the diagonals of a pentagon. These signs cannot agree. Indeed, the five-valent orbital graph is connected: a representative of the five-point suborbit lies outside D_5, and D_5 is the only proper subgroup of A_5 strictly containing C_5. Equal positive signs would make the graph two disjoint copies of K_6. Equal negative signs would give, after switching, $K_{6,6}$ minus the antipodal matching. Its unique bipartition would be preserved by A_5, which has no quotient of order two, contradicting transitivity on Ω.

Thus side and diagonal signs are opposite. After relabelling the pentagon and, if necessary, exchanging the two orbitals, we obtain the following gauge.

Triangle holonomy. For concreteness, one such gauge is

$$B = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & -1 & -1 \\ 1 & 1 & 0 & -1 & 1 & -1 \\ 1 & 1 & -1 & 0 & -1 & 1 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & -1 & 1 & 1 & 0 \end{pmatrix}.$$

The pentagon gives $B^2 = 5I$ directly. The dot product of row zero with any other row is zero because the four remaining entries split as two plus and two minus. For two nonzero rows, the contribution through vertex zero is 1, while the three remaining pentagon vertices contribute -1, whether the two indices are adjacent or diagonal. Thus every off-diagonal entry of B^2 vanishes, while each diagonal entry is five. Since $A - A'$ vanishes on the fibre-even lattice and acts as $2B$ on L^-, this also gives the full orbital identity

$$(A - A')^2 = 10(I - R).$$

It also recovers the original intersection count. For a nonantipodal pair, two of the four remaining axes have equal signs and two have opposite signs; each axis has two lifts. Hence the eight intermediate points contribute four positive and four negative products.

Its triangle products are switching-invariant. They are constant on the two A_5-orbits of three-subsets, take opposite values on complementary triples, and are not constant; after choosing the overall sign of C they are exactly the coefficients c_{ijk}. For example,

$$c_{012} = B_{01}B_{12}B_{20} = 1, \qquad c_{345} = B_{34}B_{45}B_{53} = -1.$$

These triples are complementary, so A_5-invariance proves the sign rule on both orbits. Conversely, the coefficients satisfy

$$c_{ijk}c_{ij\ell}c_{ik\ell}c_{jk\ell} = 1 \qquad (i < j < k < \ell).$$

Choose a temporary gauge with $B_{0i} = 1$. Then $B_{ij} = c_{0ij}$, and the four-point identity recovers all remaining triangle products. Thus the cubic recovers the switching class of B. Moreover, for $i \neq j$,

$$\sum_{k \neq i,j} c_{ijk} = B_{ij} \sum_k B_{ik}B_{kj} = B_{ij}(B^2)_{ij} = 0.$$

Summing these equations gives the signed one-index and total sums as zero. These are exactly the signed moments below degree three, and they imply $C(x+t\mathbf{1}) = C(x)$.

The implication reverses. Once the four-point identities have reconstructed the switching class,

$$(B^2)_{ij} = B_{ij} \sum_{k\neq i,j} c_{ijk} \qquad (i \neq j),$$

while every diagonal entry of B^2 is five. Thus signed pair balance is equivalent to $B^2 = 5I$. In the gauge $B_{0i} = 1$, the five equations $(B^2)_{0i} = 0$ say that the positive edges on the remaining five vertices form a 2-regular graph, hence a pentagon. The twelve labeled pentagons form one class up to relabeling, proving uniqueness.

The determinant pencil. For a triple $S = \{i, j, k\}$, its principal minor is

$$\det B_S = 2B_{ij}B_{jk}B_{ki} = 2c_{ijk}.$$

The identity $B^2 = 5I$ gives $B^{-1} = B/5$ and $\det B = -125$. Jacobi's complementary-minor identity then gives principal-minor values 5 in size four and 0 in size five, while complementary size-three minors have opposite signs. Expanding along the diagonal variables gives the displayed determinant formula and its homogeneous odd-part identity. In particular, the same quadratic identity that defines the golden operator also recovers support complementation.

The node frame. It remains to see that the cubic itself recovers the six axes. We use the determinantal duality behind the golden operator. Put $E = \mathbf{Q}(\sqrt{5})$, $t_\pm = (1 \pm \sqrt{5})/2$, and

$$U(t) = \begin{pmatrix} t & t & -1 \\ t & 1 & -t \\ 1 & t & -t \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The displayed conference identity gives $BU(t_\pm) = \pm\sqrt{5}\,U(t_\pm)$ and $U(t_-)^\mathsf{T} U(t_+) = 0$; write $V_\pm$ for the respective column spaces. Thus, for $D_x = \operatorname{diag}(x_0, \dots, x_5)$,

$$\Phi_x = U(t_-)^\mathsf{T} D_x U(t_+)$$

is a matrix for the cross-golden block and is unchanged by $x \mapsto x + t\mathbf{1}$. Direct expansion of this 3×3 determinant gives the exact identity

$$\det(\Phi_x) = -C(x).$$

The nonzero A_5-map $x \mapsto \Phi_x$ has irreducible source V_5, so it identifies the augmentation module with the five-dimensional summand in

$$\operatorname{Hom}_E(V_+, V_-) \cong V_4 \oplus V_5;$$

this is the standard tensor-product rule $V_3 \otimes V_{3'} = V_4 \oplus V_5$. Using the invariant forms, regard $\Phi \in \operatorname{Hom}_E(V_+, V_-)$ and $\Psi \in \operatorname{Hom}_E(V_-, V_+)$. Pair these spaces by the invariant perfect pairing

$$\langle \Psi, \Phi \rangle = \operatorname{tr}(\Psi\Phi),$$

Its annihilator has dimension four and is the complementary V_4, denoted Λ. In the displayed eigenbases, solving the six coefficient equations $\operatorname{tr}(\Psi\Phi_x) = 0$ gives the four-dimensional solution space

$$\Psi_z = \begin{pmatrix} 0 & a(z) & b(z) \\ z_0 & 0 & z_1 \\ z_2 & z_3 & 0 \end{pmatrix}, \qquad z = (z_0, z_1, z_2, z_3),$$

where

$$a(z) = (\sqrt{5} - 1)z_0 + \frac{3 - \sqrt{5}}{2} z_1 + \frac{\sqrt{5} - 5}{2} z_2 + (\sqrt{5} - 1)z_3,$$
$$b(z) = \frac{\sqrt{5} - 5}{2} z_0 + (1 - \sqrt{5})z_1 + (\sqrt{5} - 1)z_2 + \frac{\sqrt{5} - 3}{2} z_3.$$

Thus this family is all of Λ, and

$$\det(\Psi_z) = a(z)z_1z_2 + b(z)z_0z_3.$$

At $z = (1, 0, 0, 1)$ the determinant is $\sqrt{5} - 4 \neq 0$. Hence determinant restricts to a nonzero A_5-invariant cubic on Λ. The character $\chi_{V_4} = (4, 0, 1, -1, -1)$ on the classes $1, 2, 3, 5A, 5B$ gives the symmetric-cube character $(20, 0, 2, 0, 0)$, hence

$$\dim(\operatorname{Sym}^3 V_4^*)^{A_5} = \frac{20 + 20 \cdot 2}{60} = 1.$$

Therefore this determinant is, up to scalar, the Clebsch diagonal cubic

$$z_1^3 + \cdots + z_5^3 = 0, \qquad z_1 + \cdots + z_5 = 0.$$

This cubic surface is smooth: at a singular point the five squares z_i^2 would be equal, but a sum of five signs cannot vanish in characteristic zero. Here the cross-golden four-space Λ and its trace orthogonal $\Lambda^\perp$ define exactly the paired determinantal hypersurfaces of Hassett and Tschinkel. Their Proposition 10 therefore applies: smoothness of the cubic surface $\mathbf{P}(\Lambda) \cap \Sigma_2$ implies that $\det(\Phi_x) = 0$ in $\mathbf{P}(\Lambda^\perp)$ has exactly six ordinary double points in linear general position [15, §3, Proposition 10].

Pair balance makes each $p_a = [\mathbf{1} - 6e_a]$ singular, and these six points already form a projective frame. They are therefore the complete singular locus. Their node type also follows directly from the same golden identity. At p_a, pair balance gives

$$\operatorname{Hess}(C)_{ij}(p_a) = \begin{cases} -6c_{aij}, & a \notin \{i, j\}, \\ 0, & a \in \{i, j\}. \end{cases}$$

The nonzero block is switching-equivalent to a five-by-five principal minor M of B. If u is the deleted row, the corresponding block decomposition of $B^2 = 5I$ gives

$$Mu = 0, \qquad M^2 = 5I - uu^\mathsf{T}.$$

Since $u^\mathsf{T}u = 5$, the second identity restricts to $M^2 = 5I$ on $u^\perp$. Thus $\ker M = \mathbf{Q}u$ and $\operatorname{rank} M = 4$. Moreover $\operatorname{tr} M = 0$, so the two square roots of 5 occur twice each and

$$\chi_M(\lambda) = \lambda(\lambda^2 - 5)^2.$$

The Hessian therefore has precisely the translation and radial directions as its kernels. Its induced form on the four-dimensional projective tangent space is nondegenerate, so every singularity is an ordinary double point.

Frame symmetry. The node frame identifies every projective cubic automorphism with a frame permutation. Let $N \leq S_6$ normalize the five matchings $\mathcal{T}$ in Figure 1. Its action on $\mathcal{T}$ is faithful, and the displayed table supplies normalizers inducing generators of A_5 and an odd permutation; hence $N \cong S_5$. The even generators preserve every c_{ijk} and the odd element reverses them all. Moreover, the identity and the five transpositions $(0\,i)$ give six distinct cubic lines. Thus the line stabilizer has order at most $720/6 = 120$ and equals N; its orientation character is the sign on $\mathcal{T}$. The cubic-line and oriented-cubic stabilizers are therefore S_5 and A_5.

The integral commutant. Finally, after extension from $\mathbf{Q}$ to $E = \mathbf{Q}(\sqrt{5})$, the odd module splits as the two nonisomorphic three-dimensional icosahedral representations:

$$L^- \otimes_{\mathbf{Z}} E \cong V_3 \oplus V_{3'}.$$

They are exchanged by the nontrivial element of $\mathrm{Gal}(E/\mathbf{Q})$, and B acts on them by the conjugate scalars $\sqrt{5}$ and $-\sqrt{5}$. Schur's lemma followed by Galois descent therefore gives

$$\mathrm{End}_{\mathbf{Q}A_5}(L^- \otimes \mathbf{Q}) = \mathbf{Q}[B] \cong E.$$

If $aI + bB$ preserves L^-, its diagonal entries force $a \in \mathbf{Z}$, while any off-diagonal unit forces $b \in \mathbf{Z}$. This proves the integral commutant statement. $\square$

Remark 8.3 (The conductor at two). The order $\mathbf{Z}[\sqrt{5}]$ has conductor two in $\mathbf{Z}[(1+\sqrt{5})/2]$. The orientation collapse is already visible without normalizing: modulo 2, all c_{ijk} become 1, while $B - I$ has rank one and square zero. This is precisely the degeneration

$$\mathbf{Z}[\sqrt{5}] \otimes \mathbb{F}_2 \cong \mathbb{F}_2[u]/(u-1)^2.$$

No ambient characteristic-zero geometry is used here.

9 Proof sources and formal cross-check

The rigidity and field-window arguments use three external mathematical inputs, while the orientation section uses one external model identification and one determinantal-duality theorem. Dye's Brianchon bound and equality classification identify the Clebsch orbit [11, §2.2–2.3, Theorem 1, pp. 275–276]; his stabilizer and associated-conic results supply the classical Clebsch geometry [11, Theorems 2–3, pp. 276–278]. The even-order oval obstruction uses the standard nucleus theorem [14, p. 177]. The upper conic-filling bound is Proposition 1.6 of Blokhuis, Brouwer, and Szőnyi [3]. Cheltsov, Tschinkel, and Zhang supply a published S_5-symmetric equation inside their classification of cubics already assumed to be six-nodal. Section 8 gives the explicit coordinate identification, but does not use that citation to transfer singular-locus completeness. That transfer instead uses the equivalence in Hassett–Tschinkel for the determinantal hypersurfaces attached to a four-space in $\mathrm{End}(V)$ and its trace orthogonal: the cubic threefold has six ordinary nodes in linear general position precisely when the paired cubic surface is smooth [15, §3, Proposition 10].

The line bound, chord-defect identity, partial-cover deduction, decoder reconstruction, support bipartition, and the orientation two-graph reconstruction are proved in the text. In particular, the conference identity gives pair balance and the Hessian ranks, while the five-matching normalizer and six-coset bound give the S_5/A_5 stabilizer distinction. The golden cross block, its four-dimensional trace complement, and the smooth Clebsch dual surface put the cubic under the cited determinantal theorem, so singular-locus completeness is structural. The paper-owned replay independently constructs the coset actions, signed operator, triangle products, diagonal pencil, cross-golden trace dual, switching class, gradient ideals, and Hessian blocks. The gradient exhaustion is independent rather than the proof of completeness. It reads no later-paper artifact:

`check_orientation_two_graph.py`.

None of the preceding manuscript conclusions is imported from the formal development. After the structural proofs are complete, the formal gate independently rederives the displayed q11 code and decoder conclusions, the causal containing-quadratic rigidity spine, the projective-to-code correspondence, the chord-defect specialization, the $q = 9$ obstruction, the small-arc moments, and the eight orientation mechanisms from the antipodal cover through the rational and integral commutants. Dye's bound and classification remain two explicit axioms. The commutant theorems instead take an explicit proposition-valued interface for the classical conjugate $3 + 3'$ decomposition, Schur's lemma, and Galois descent; Lean checks golden equivariance, the reverse containment, and the diagonal/off-diagonal integral test. The generated q11 rows are finite leaves internal to this redundant cross-check; they are not premises of the orbit–stabilizer or incidence proofs above. The companion searches lie outside this gate. The reusable Lean development is distributed in `finitegeom`, while the generated q11 tables and aggregate Paper I gate are in `the q11 package`. The base library's version-independent archival locator is the Zenodo concept DOI `10.5281/zenodo.21650878`. The formalization uses Lean 4.32.0-rc1. Its pinned artifacts are

certificate-package commit: `09d8e174880e7370966da788da3c5d303df8af4f`
base-library commit: `570086982b26075a71a331a81bb1b519e9a27e7f`
aggregate gate: `RelativeConicArcs/Gates/ClebschRigidityTrust.lean`
Mathlib commit: `571b8a8e54219b4d393f75f4b8653fac08197fcc`.

The SHA-256 digest of `verification/checker_outputs.json` is

`cd4386336e06e11c9e2aa3228c18e828ad216bc78fd05f7387502ab6a06d0e04`.

The SHA-256 digest of `RelativeConicArcs/Gates/ClebschRigidityTrust.lean` is

`c5d532dbd79dcb2eef602ced85105b72943a0a1af05de11c3c008c1ed9a1d747`.

This formalization is an independent cross-check, not a proof dependency of the manuscript. Its recorded axiom audit lists the standard logical axioms and exactly the two Dye axioms above for the rigidity implication; the gate uses neither admitted declarations nor native execution. Because the classical $3 + 3'$ input is a theorem parameter rather than a global axiom, the trust manifest records that interface beside the two commutant terminals.

10 Conclusion

For every fixed arc size, the partial-cover bound limits the field orders that can admit conic filling. Together with the universal defect identity and passant counts, it gives

$$2k - 3 \le q \le \frac{k(k-1)+3}{3}.$$

The Clebsch code realizes a different kind of sharpness: covering-radius data recover geometry that was not built into the code, forcing the Clebsch hexagon and hence its A_5 symmetry. A line bound and the defect identity reduce that rigidity to Dye's equality case.

More generally, which odd prime powers in the quadratically widening window admit conic-filling arcs? A second problem is to connect fixed-k rigidity with the large-arc conic-forcing regime. Both ask how far a syndrome locus can reconstruct the projective system that produced it.

The same questions have useful algebraic and spectral forms. Passant conditions become clique problems in conic association schemes, while the decoder results ask when coset-leader support data alone reconstruct a projective code. For the Clebsch code, that support data also reconstructs the switching class of the golden continuation operator: the support cubic and the twelve-point syndrome locus retain the same orientation two-graph. The determinant pencil makes the bridge visible: its quadratic identity $B^2 = 5I$ controls every symmetric layer, while triangle holonomy is exactly the remaining cubic term.

AI assistance disclosure

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